Viscan Voice Dental Assistant

Voice-first report retrieval and dictation drafting for dental imaging

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github.com/DrEnosh/Vi-scan
Python · FastAPI · Groq (gpt-oss-120b, Whisper)

DESIGN PRINCIPLE

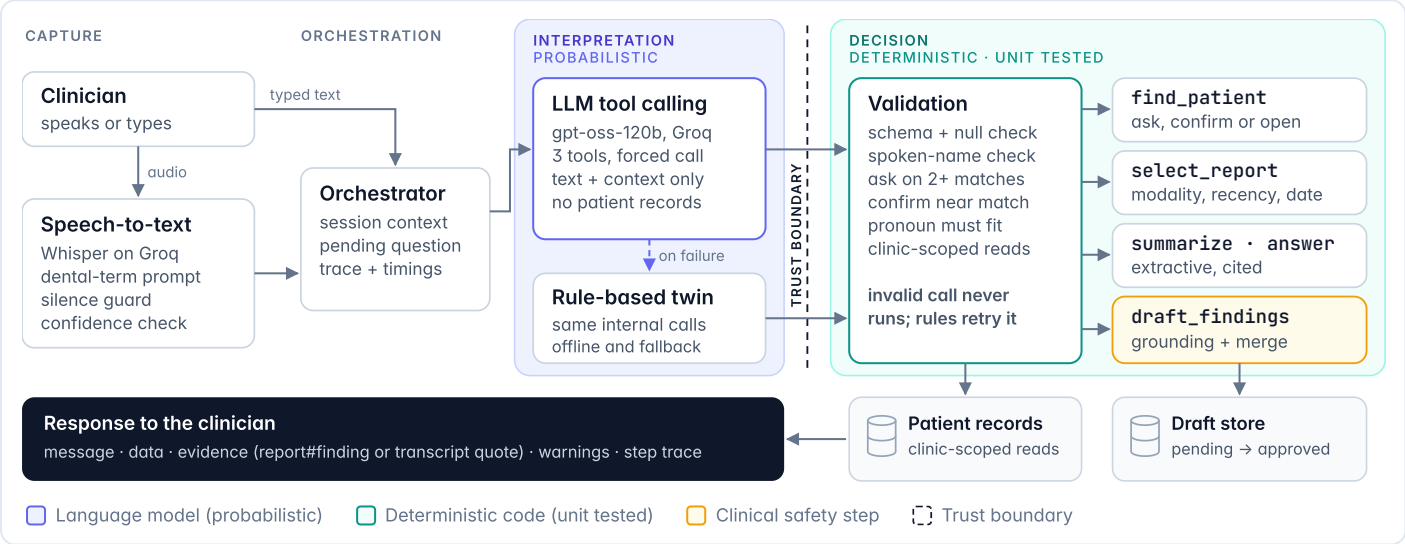
The model interprets. Code decides.

A model output is only a proposal. Patient identity, report selection and every draft finding are checked by tested, deterministic code before the clinician sees them. Nothing enters a report that was not said.

≈1.3 s	9 / 9	62 + 4	3	$0.0003
Spoken command, server side (live mean)	Correct tool calls in the live run	Automated tests + live API tests, all passing	Tools the model can call; everything else is code	Per spoken command at paid API rates

01 System architecture

Arrows show the request path; the dashed line is the trust boundary



02 Request flow

1 Voice to text
The browser records audio and posts it to `/api/assistant/voice`. Whisper transcribes it with a dental-term prompt; known silence hallucinations are rejected and low-confidence transcripts flagged. Typed text skips this step.

2 Interpretation
The utterance and a one-line context go to the model, which must call exactly one of three tools. It never sees patient records. On a timeout, rate limit or unusable reply, the rule-based twin makes the same call.

3 Validation and tools
Arguments are schema-checked and an invalid call never runs. Code restores a dropped spoken name, resolves the patient (ask, confirm or open), checks pronouns, enforces clinic scope and selects the report.

4 Report generation and evidence
Each dictated finding must quote the transcript and name the finding; tooth and negation are re-derived. A parser adds missed findings. Drafts stay pending until approved, and summaries cite `report_id#finding_id`.

03 Production considerations

CONCERN	IN THIS PROOF OF CONCEPT	NEXT STEP FOR PRODUCTION
Privacy	The model sees the utterance and a one-line context only; the audit log keeps IDs, not clinical text. Synthetic data.	Provider DPA or in-region self-hosting (DPDP Act), zero retention, encrypted drafts, PHI-free logs.
Authorization	Clinic-scoped reads; out-of-scope patients look non-existent; sessions bound to one clinician; drafts private.	OIDC tokens in place of the demo header, role-based sign-off, break-glass access logging.
Latency	Live: STT 0.35–0.48 s, model 0.59–1.61 s, code ≈1 ms. An 8 s model timeout falls back to rules at once.	Streaming STT; rules-first routing so the model handles dictation and unclear phrasing only.
Cost	Free on Groq's free tier, at about 860 input tokens per model call.	About $0.0003 per spoken command at paid rates; self-host Whisper and an open-weight model at volume.
Failure handling	Model error or HTTP 429: rule fallback. STT failure: clear 4xx. Input limits; no stack traces to clients.	Retry budgets, a circuit breaker, and alerts on rejection and clarification rates.

All patient data in this project is synthetic. Appendix A and B (pages 2–3) give lifecycle, safety and verification detail.

APPENDIX A

Request lifecycles and safety controls

Three requests traced through the system as they ran against the live APIs, followed by the safety controls each clinical risk maps to and the tests that hold them in place.

A1 Request lifecycles

Colour shows who acts at each step: clinician, model, code, outcome

Retrieval with an ambiguous name

one model call, then code only

CLINICIAN	SPEECH-TO-TEXT	MODEL	CODE	CLINICIAN	OUTCOME
"Open Priya's latest OPG report"	Transcribed in ≈0.4 s; passes the silence guard	`report_request`: open, "Priya", OPG, latest	Two patients match, so it asks which (ID, age, last scan)	*"Priya Sharma"*: resolved in code, no model call	R-1003 opened, with a look-alike alert (Priya Shetty)

Dictation to a structured draft

model recall, code precision

CLINICIAN	MODEL	GROUNDING	PARSER MERGE	DRAFT	CLINICIAN
Dictates four sentences, including *"possible caries on 46"*	`draft_findings`: three findings, each with a quote	Quote present and names the finding; tooth and negation re-derived	Adds one missed finding: canal proximity at 48	4 findings, 1 flagged, the remark kept verbatim	Edits, links the patient and approves; the draft is frozen

Follow-up with session context

context is used only when it fits

CLINICIAN	MODEL	CODE	OUTCOME	NEXT QUESTION
"Summarize her report"	`report_request`: summarize, pronoun female	Open patient P001 is female, so context is accepted	Extractive summary citing `R-1003#F1` to `F5`	*"Any caries?"* Not in R-1003; found in R-1002 and R-1001

A2 Safety controls

Each control is enforced in code and covered by tests

RISK	CONTROL	ENFORCED IN	VERIFIED BY
Wrong patient: shared name	Ask on two or more matches, confirm near-matches, and flag look-alike names with ID and age.	`match_patients`	`test_ambiguous_identity (9)`
Wrong patient: model slip	A patient name or ID from the model must appear in the utterance, or it is discarded.	`orchestrator`	`test_ambiguous_identity`
Wrong patient: context	A pronoun must fit the open patient; a spoken name beats context; an unknown name is never swapped for the open patient without a yes.	`orchestrator`	`test_followup (8)`
Invented or altered findings	A verbatim quote that names the finding is required; tooth, negation and hedging are re-derived from the words.	`ground_llm_findings`	`test_unsupported_claims (11)`
Guessed tooth	"Lower third molar" stays unresolved as 38 or 48, and approval is blocked until a tooth is picked.	`drafts.approve`	`test_api (7)`
Silence read as a command	Whisper silence phrases rejected, no-speech segments dropped, low-confidence transcripts flagged.	`OpenAICompatSTT`	`test_providers (18), live run`
Malformed tool call	Pydantic schemas with nulls treated as absent; an invalid call is re-interpreted, never executed.	`validate_tool_call`	`test_unsupported_claims`
Cross-clinic access	Every read is clinic-scoped; out-of-scope records return "not found", so existence is not leaked.	`Repository`	`test_retrieval (9)`
Unreviewed draft	No impression or advice is generated; drafts stay pending until approved, and approved drafts are frozen.	`tools/drafts`	`test_api`
Data exposure to the model	The model gets the utterance and a one-line context only; the audit log records IDs, not text.	`interpreter, _audit`	`request-shape test`

APPENDIX B

Design decisions, live verification and roadmap

The choices that shape the system, what running it against the real APIs showed, and the path from proof of concept to a clinical pilot.

B1 Key design decisions

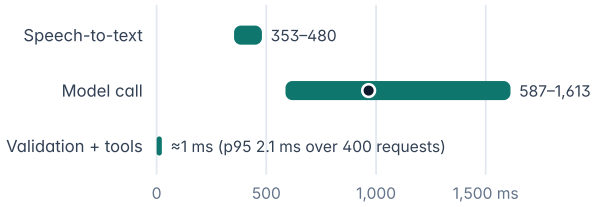
DECISION	WHY	TRADE-OFF ACCEPTED
The model proposes; code executes	Identity, scope and evidence rules stay deterministic, auditable and tested.	Two layers to maintain.
Rule-based twin, same internal calls	Offline mode, deterministic tests, and a ≈1 ms fallback when the model fails.	Narrower phrasing coverage than the model.
Extractive summaries and answers	Every line traces back to a report field through `report_id#finding_id`.	Less fluent than generated prose.
Grounding check plus parser merge	The model brings recall, code brings precision; nothing enters a draft unquoted.	A 23-type vocabulary; the rest is kept verbatim.
Three tools and a static prompt	About 860 tokens per call, cacheable, and within free-tier limits.	One report tool with an action field.
Hosted Whisper with a dental prompt	0.35–0.48 s per clip, with terms such as OPG and mesioangular recognised.	Real audio needs a DPA or self-hosting.

B2 Live verification

Groq free tier, Windows, 13 October 2026 · full log in docs/LIVE_RUN.md

Latency by stage

Range per request in the live run; the dot marks the mean of 9 model calls (966 ms)



- **9 of 9 tool calls correct**: six typed turns and three spoken clips
- **4 of 4 live API tests passed**; the 62 offline tests need no key
- **Whisper matched every clip**; "tooth 4-6" resolved to tooth 46
- **Silence rejected**: a silent clip read as "Thank you." never reached the model
- **No guessing**: "Priya" still produced a clarification

B3 What live testing changed

Each issue now has a fix and a regression test

FOUND AGAINST THE REAL APIS	FIX
Groq rejected tool calls with `null` arguments (HTTP 400)	Optional fields made nullable; nulls dropped before validation.
Whisper transcribed 0.5 s of silence as "Thank you."	Silence-phrase guard plus no-speech and low-confidence checks.
Free tier allowed ≈6 calls a minute at 1,300 tokens each	Merged tools and a static prompt: ≈860 tokens per call, instant fallback on 429.
The model skipped one dictated finding	Grounded parser findings merged in; misattributed qualifiers dropped.
The model once read "Priya's" as a pronoun	Spoken-name cross-check before session context is used.

B4 Roadmap to a clinical pilot

PHASE 1

Pilot hardening

- OIDC sign-in and role-based approval
- Postgres for sessions, drafts and versions
- Immutable audit trail

PHASE 2

Clinical quality

- Evaluation set: Indian-English accents, code-switching, clinic noise
- Track dental-term WER and extraction precision and recall

PHASE 3

Scale and integration

- Streaming STT, rules-first routing
- In-region self-hosted models
- PACS/RIS via FHIR DiagnosticReport and DICOM SR